BirdCLEF-2026 — Weekly Progress

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- Experiment 005 completed — token context shaping validates rectangular patches
- Experiment 006 results — SSAMBA trained from scratch on BirdSet HSN: 0.086 cmAP
- 5 new concept docs exploring directions to close the gap
- 2 new experiment proposals: 007 (token-level SED), 008 (multi-signal SED)

Exp 005 — Token Context Shaping

- **Hypothesis**: rectangular patches (128xT) enable practical 30–60s SSAMBA inference
- 11 patch configs tested across tiny and small models, 30s and 60s audio
- Mamba handles 12k+ tokens at 84ms — speed is not the bottleneck
- **Verdict**: confirmed — rectangular patches unlock long-context audio

Exp 005 — GPU Benchmark (Tiny, 30s)

Config	Patch	Tokens	ms/fwd	tok/s
baseline-16x16	16×16	1,496	65.4	22,881
fullband-128x2	128×2	1,500	28.3	52,964
framelevel-128x1	128×1	3,000	29.2	102,607
halfframe-64x1	64×1	6,000	43.1	139,150
quarterframe-32x1	32×1	12,000	84.4	142,227
eighthframe-8x1	8×1	48,000	490.5	97,858
fine-4x1	4×1	96,000	2,010	47,751

Exp 005 — Scaling Regimes

Three regimes identified:

1. **Floor** (<3k tokens): fixed ~28ms overhead dominates
2. **Linear** (6k–24k): SSM compute dominates, peak 142k tok/s
3. **Superlinear** (>48k): activations exceed GPU L2 cache

O(n) verification — 30s vs 60s ratio:

Config	30s	60s	Ratio
framelevel	29ms	51ms	1.7×
halfframe	43ms	80ms	1.9×
quarterframe	84ms	165ms	2.0×

baseline-16×16 anomaly: **2.3× slower** than equivalent rectangular config

Exp 005 — Practical Implications

- **Best configs for BirdCLEF**:
  - framelevel-128×1 (3k tokens, 29ms) — full frequency, 10ms resolution
  - quarterframe-32×1 (12k tokens, 84ms) — peak throughput, 4 freq bands
- **Memory for training**: 3k tokens → ~90 samples/batch on 24GB; 12k → ~33/batch
- **Conclusion**: speed and memory are solved — learning quality is the bottleneck

Exp 005 — ONNX CPU Benchmark (60s audio)

Tiny model (192d), 8 cores — Kaggle: 700 × 60s clips, 90-min CPU-only

Config	Tokens	ms/60s	700 clips	Budget
fullband-128×2	3,000	1,985	23 min	26%
framelevel-128×1	6,000	2,723	32 min	35%
halfframe-64×1	12,000	5,028	59 min	65%
quarterframe-32×1	24,000	11,181	130 min	145%

- **Backend**: mamba-onnx custom op (Zig, AVX2)
- **Max feasible on 8 cores**: 12k tokens at 65% budget
- Kaggle has 4 cores — safe limit likely **6k tokens** (framelevel-128×1)
- INT8 quantization not yet tested — could push limit higher

Exp 006 — SSAMBA BirdSet HSN Setup

- **Model**: SSAMBA Tiny (192d, 24 layers, ~7M params)
- **Patch config**: framelevel-128×1 → 3,000 tokens per 30s
- **Data**: BirdSet HSN (21 species), 5,460 training recordings
- **Validation**: 12,000 five-second soundscape clips
- **Training**: 100 epochs max, lr=5e-4, warmup=10, label_smoothing=0.1, bf16-mixed
- **Augmentation**: time masking (300 frames), frequency masking (24 bins)

Exp 006 — Results

Baseline comparison:

Model	Pre-trained?	cmAP
ConvNeXT	ImageNet	0.48
ViT	No	0.164
SSAMBA Tiny	**No**	**0.086**

Training trajectory (version 2):

Epoch	val/cmAP	val/AUROC
0	0.026	0.464
10	0.053	0.500
15	0.079	0.509
18	**0.086**	**0.522**
21	0.081	0.508

Best checkpoint: epoch 18 (early stopping ~epoch 30)

Exp 006 — Diagnosing the Gap

- The **0.086 vs 0.48 gap** is too large for a single factor
- **Ranked causes**:
  1. **No pre-training** (dominant) — ConvNeXT uses ImageNet features
  2. **Zero negatives in training** — all chunks labeled positive, but validation is ~55% empty
  3. **Architecture suitability** — from-scratch ViT still gets 0.164
  4. **Training dynamics** — lr, augmentation, batch size need tuning
- **Key insight**: forward pass is 29ms — speed is fine; **learning quality** is the bottleneck

Concept — Unsupervised Event Detection

- **Problem**: all training chunks labeled positive (no negatives)
- **Solution**: bambird detects vocalization events in XenoCanto recordings
  - Chunks **without** detected events become natural negatives
  - Lightweight, proven approach (BirdSet uses it internally)
  - One-time preprocessing cost, cached with spectrograms
- **Impact**: directly addresses the zero-negatives problem from exp 006

Concept — Weak Supervision Label Fusion

- **Combine multiple noisy sources**: bambird, ConvNeXT, BirdNET, energy detector
- **FlyingSquid triplet method**:
  - Closed-form accuracy estimation — no ground truth needed
  - Produces calibrated soft labels with per-source accuracy weights
  - ~30 lines numpy, instant computation, no hyperparameters
- **Result**: better labels than any single source alone

Concept — Learned Frequency Tokenizer

- **Problem**: Linear(128, 192) is too weak — no nonlinearity, no hierarchy
- **Solution**: LEAF (learnable Gabor filterbank) + frequency attention + Mamba
  - LEAF: 30K params, learns filterbank from data
  - Frequency attention: 1–2 transformer layers over 128 freq tokens
- **Quick diagnostic first**: swap Linear for 3-layer MLP to test if depth matters
- **Modular SSL**: pre-train frequency encoder and Mamba backbone separately

Concept — Pre-trained Mamba Transfer

- Load AudioSet pre-trained SSAMBA weights
- Re-initialize **only** patch embedding (~1% of params)
- Mamba layers transfer naturally — state dynamics are about audio, not tokenization

Length extension — Mamba vs Transformer:

Property	Mamba	Transformer
Pos. encoding	None	Required
Train 10s $\rightarrow$ eval 60s	Native	Degrades
Memory scaling	$O(1)$	$O(n^2)$
Speed scaling	$O(n)$	$O(n^2)$

Concept — Prompted Sound Event Detection

- **Reference audio clip as Mamba prefix** $\rightarrow$ per-token binary output
- Solves: domain gap, low-resource species, zero-shot detection
- Mamba as **learned matched filter**: $O(n+m)$ vs $O(n \cdot m)$ cross-attention
- **3-stage training**:
  1. SSL pre-training (masked token prediction)
  2. Episode-based prompted SED (within-recording, cross-recording, negatives)
  3. Soundscape adaptation with pseudo-labels

Next Week Plan

- **Pre-trained SSAMBA Tiny baseline** on BirdSet HSN
  - Load AudioSet weights, re-initialize patch embedding (128×1)
  - Freeze backbone initially, then unfreeze for end-to-end fine-tuning
- **Target**: >0.15 cmAP (vs 0.086 from scratch)
- **Training**: 10s windows $\rightarrow$ evaluate on 60s (length-agnostic recurrence)
- Begin exploring prompted SED direction

Experiment Status Summary

ID	Name	Status	Key Result
005	Token Context Shaping	Complete	12k tokens feasible, O(n) confirmed
006	SSAMBA BirdSet HSN	Complete	0.086 cmAP — needs pre-training
007	Token-Level SED	Proposed	bambird temporal labels
008	Multi-Signal SED	Proposed	distillation + bambird + SSL